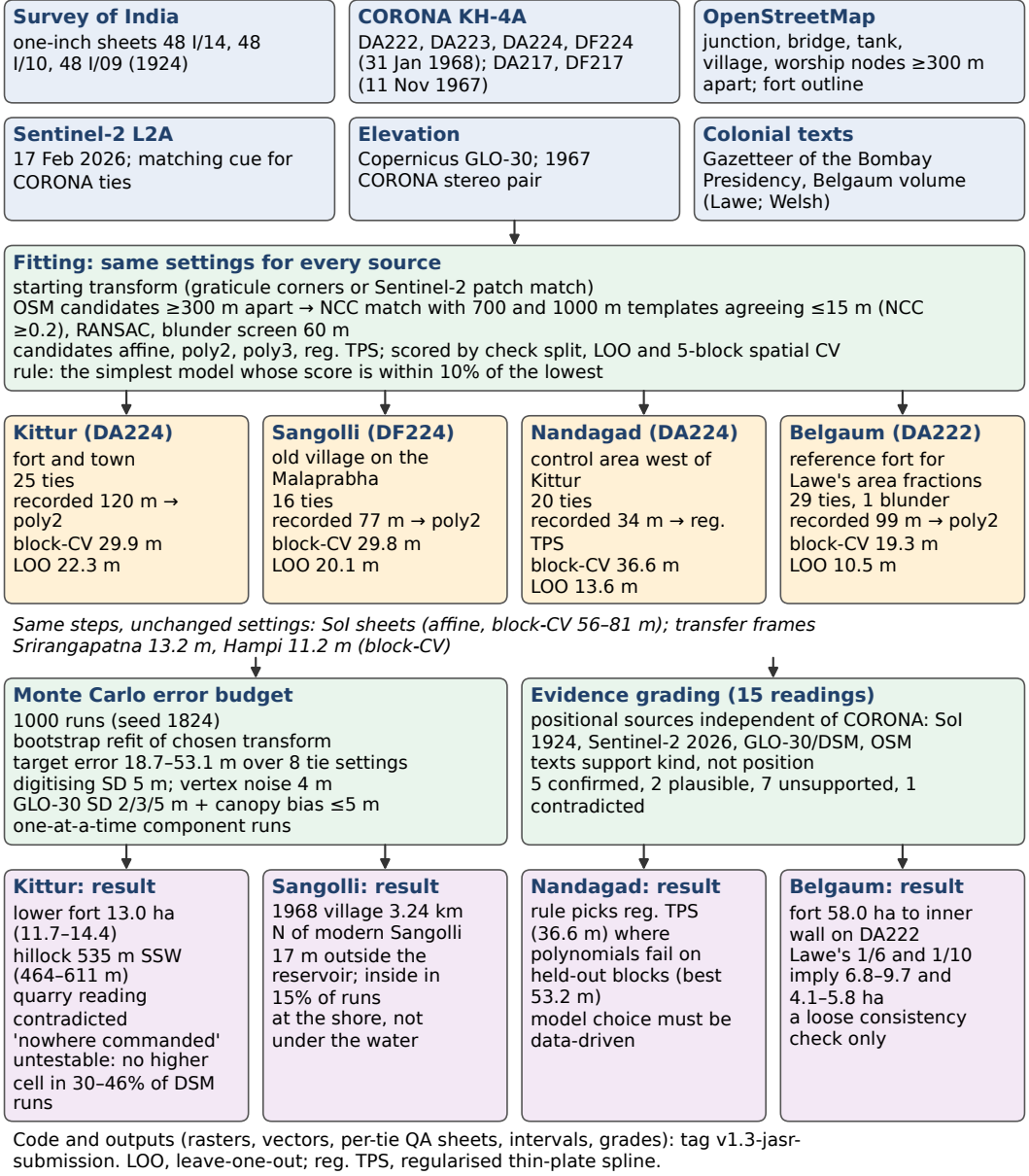
Survey of India
one-inch sheets 48 I/14, 48 I/10, 48 I/09 (1924)
CORONA KH-4A
DA222, DA223, DA224, DF224 (31 Jan 1968); DA217, DF217 (11 Nov 1967)
OpenStreetMap
junction, bridge, tank, village, worship nodes ≥300 m apart; fort outline
Sentinel-2 L2A
17 Feb 2026; matching cue for CORONA ties
Elevation
Copernicus GLO-30; 1967 CORONA stereo pair
Colonial texts
Gazetteer of the Bombay Presidency, Belgaum volume (Lawe; Welsh)
Fitting: same settings for every source
starting transform (graticule corners or Sentinel-2 patch match)
OSM candidates ≥300 m apart → NCC match with 700 and 1000 m templates agreeing ≤15 m (NCC ≥0.2), RANSAC, blunder screen 60 m
candidates affine, poly2, poly3, reg. TPS; scored by check split, LOO and 5-block spatial CV
rule: the simplest model whose score is within 10% of the lowest
Kittur (DA224)
fort and town
25 ties
recorded 120 m → poly2
block-CV 29.9 m
LOO 22.3 m
Sangolli (DF224)
old village on the Malaprabha
16 ties
recorded 77 m → poly2
block-CV 29.8 m
LOO 20.1 m
Nandagad (DA224)
control area west of Kittur
20 ties
recorded 34 m → reg. TPS
block-CV 36.6 m
LOO 13.6 m
Belgaum (DA222)
reference fort for Lawe's area fractions
29 ties, 1 blunder
recorded 99 m → poly2
block-CV 19.3 m
LOO 10.5 m
Same steps, unchanged settings: SoI sheets (affine, block-CV 56–81 m); transfer frames Srirangapatna 13.2 m, Hampi 11.2 m (block-CV)
Monte Carlo error budget
1000 runs (seed 1824)
bootstrap refit of chosen transform
target error 18.7–53.1 m over 8 tie settings
digitising SD 5 m; vertex noise 4 m
GLO-30 SD 2/3/5 m + canopy bias ≤5 m
one-at-a-time component runs
Evidence grading (15 readings)
positional sources independent of CORONA: SoI 1924, Sentinel-2 2026, GLO-30/DSM, OSM
texts support kind, not position
5 confirmed, 2 plausible, 7 unsupported, 1 contradicted
Kittur: result
lower fort 13.0 ha (11.7–14.4)
hillock 535 m SSW (464–611 m)
quarry reading contradicted
'nowhere commanded' untestable: no higher cell in 30–46% of DSM runs
Sangolli: result
1968 village 3.24 km N of modern Sangolli
17 m outside the reservoir; inside in 15% of runs
at the shore, not under the water
Nandagad: result
rule picks reg. TPS (36.6 m) where polynomials fail on held-out blocks (best 53.2 m)
model choice must be data-driven
Belgaum: result
fort 58.0 ha to inner wall on DA222
Lawe's 1/6 and 1/10 imply 6.8–9.7 and 4.1–5.8 ha
a loose consistency check only
Code and outputs (rasters, vectors, per-tie QA sheets, intervals, grades): tag v1.3-jasr-submission. LOO, leave-one-out; reg. TPS, regularised thin-plate spline.